Data Communication & Computer Networks — Topic 1: Introduction of Data Communication

Practice Test Booklet — 144 MCQs (2-column, without solutions) · Easy + Medium + Hard mix · Har question ke saath **Source** · Print ke liye · Reference: Forouzan + India-level & worldwide exam papers (GATE/NET/KVS/EMRS/DSSSB/SSC/RRB/IBPS) · Prepared for: RPSC Programmer • ACP • RSMSSB IA • SA • DD • BCI • SCI • REET • RAS • DSSSB • KVS • NVS • HTET • SSC CGL/CHSL • RRB • IBPS • UPSC & all Rajasthan + Central Govt. Exams (Computer Awareness)

Section A - DC Components & Protocol Basics (Message, Sender, Receiver, Transmission Medium, Protocol)

Q1. The physical path between the sender and receiver is called:

Sender और receiver के बीच का physical path _____ कहलाता है:

(A) Link / link (link) **(B)** Node / node (node)
(C) Segment / सेगमेंट (segment) **(D)** Socket / सॉकेट (socket)

[Source: Repo: BCI_SCI Subject Test-7 (UNIT TEST-2 BCI/SCI), Q20] · Easy

Q2. Identify the five components of a data communication system.

Data communication system के पाँच components की पहचान कीजिए।

(A) Message, Sender, Receiver, Transmission medium, Protocol **(B)** Message, Router, Switch, Cable, Software
(C) Data, Sender, Receiver, Hub, TCP/IP **(D)** Message, Sender, Receiver, Software, Hardware

[Source: Forouzan Ch1 (Q1-1) — Practice Set] · Medium

Q3. The _____ is the physical path over which a message travels from sender to receiver.

_____ वह physical path है जिस पर message sender से receiver तक जाता है।

(A) Path / पथ **(B)** Medium
(C) Protocol **(D)** Route / रूट (path)

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q35] · Medium

Q4. The information to be communicated in a data communication system is called the _____.

Data communication system में communicate की जाने वाली information _____ कहलाती है।

(A) Medium **(B)** Protocol
(C) Message / message (message) **(D)** Transmission / संचरण (transmission)

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q38] · Medium

Q5. Which of the following shares the data of two computers?

निम्नलिखित में से कौन दो computers का data share (साझा) करता है?

(A) Library / पुस्तकालय **(B)** Network
(C) Grouping / समूह (grouping) **(D)** Integrated system / एकीकृत प्रणाली

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q13] · Medium

Q6. _____ refers to two characteristics: when data should be sent and how fast it can be sent.

_____ दो characteristics को बताता है: data कब भेजा जाए और कितनी तेज़ी से भेजा जा सकता है।

(A) Semantics / अर्थ (semantics) **(B)** Syntax / संरचना (syntax)
(C) Timing / समय-निर्धारण (timing) **(D)** none of the above / इनमें से कोई नहीं

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q44] · Medium

Q7. What are the main elements of a protocol? 1. Syntax 2. Semantics 3. Timing

किसी protocol के मुख्य तत्व (elements) क्या हैं? 1. Syntax 2. Semantics 3. Timing

(A) 1, 2 only / सिर्फ 1 और 2 **(B)** 1, 3 only / सिर्फ 1 और 3
(C) 2, 3 only / सिर्फ 2 और 3 **(D)** 1, 2, 3 only / 1, 2 और 3 सभी

[Source: Repo: CN Topic Test-02 (UNIT TEST-2), Q3] · Medium

Q8. Why are protocols needed in data communication?

Data communication में protocols की आवश्यकता क्यों होती है?

(A) To enable communication between different types of systems by providing common rules / अलग-अलग प्रकार की systems के बीच communication के लिए common rules देना
(B) To increase the processing speed of the CPU / CPU की processing speed बढ़ाने के लिए
(C) To convert every message into a common language / हर message को एक common language में बदलने के लिए **(D)** Protocols are optional — devices can communicate without rules / Protocols optional हैं — बिना rules के भी devices communicate कर सकती हैं

[Source: Forouzan Ch1 (Q1-11) — Practice Set] · Medium

Q9. A set of rules that governs data communication is called a:

Data communication को regulate (नियंत्रित) करने वाले rules का set _____ कहलाता है:

(A) rule / नियम (rule) **(B)** medium / medium (medium)
(C) Link / link (link) **(D)** Protocol / protocol (protocol)

[Source: Repo: CN Topic Test-03 (UNIT TEST-3), Q15] · Medium

Q10. Consider: (1) Twisted pair cables (2) Microwaves and satellite signals (3) Repeaters (4) Analog transmissions (5) Fiber optics. Which of the above is considered as signal transmission medium in data communications?

विचार करें: (1) Twisted pair cables (2) Microwaves और satellite signals (3) Repeaters (4) Analog transmissions (5) Fiber optics। उपरोक्त में से data communications में signal transmission medium के रूप में क्या माना जाता है?

(A) (1) and (5) / (1) और (5) **(B)** (1) and (2) / (1) और (2)
(C) (1), (2) and (5) / (1), (2) और (5) **(D)** (1), (2), (3) and (5) / (1), (2), (3) और (5)

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q10] · Medium

Q11. Data communication is the exchange of _____ between two devices via some form of transmission medium.

Data communication = do devices के बीच _____ का exchange, किसी transmission medium के through।

(A) only text messages / सिर्फ text messages **(B)** data (the message) - text, numbers, images, audio or video / data (message) - text, numbers, images, audio या video
(C) only electrical signals / सिर्फ electrical signals **(D)** only hardware devices / सिर्फ hardware devices

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

Q12. The device that sends the data message in a data communication system is called the _____.

Data communication system में जो device data message भेजता है, use _____ कहते हैं।

(A) Receiver / Receiver (प्राप्तकर्ता) **(B)** Sender / Sender (प्रेषक)
(C) Protocol / Protocol (protocol) **(D)** Hub / Hub (hub)

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

Q13. Which of the following best lists the popular forms of information (message) in data communication?

Data communication में information (message) के popular forms को सबसे achhe से कौन बताता है?

(A) Only text and numbers / सिर्फ text और numbers **(B)** Only audio and video / सिर्फ audio और video
(C) Text, numbers, images, audio and video / text, numbers, images, audio और video **(D)** Only binary digits / सिर्फ binary digits

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

Q14. ARPANET stands for:

ARPANET का पूरा नाम है:

(A) Advanced Research Projects Agency Network **(B)** Advanced Recheck Projects Agency Internet
(C) American Research Projects Agency Network **(D)** Advanced Research Protocol Agency Network

[Source: IBPS/SBI computer-awareness banks (via tpointtech & meritnotes)] · Medium

Q15. In a protocol, the meaning of each section of the message is called:

एक protocol में message के हर भाग के अर्थ को क्या कहते हैं?

(A) Syntax **(B)** Semantics
(C) Timing **(D)** Encoding

[Source: Practice — original (textbook-based)] · Medium

Q16. Coaxial cable and fiber optic cable are examples of:

Coaxial cable और fiber optic cable किसके उदाहरण हैं?

(A) Guided media **(B)** Unguided media
(C) Wireless media **(D)** Simplex media

[Source: Practice — original (textbook-based)] · Medium

Q17. In a computer network, nodes are:

Computer network में nodes होते हैं:

(A) the computer that originates the data / जो computer data को originate (shurua) करता है **(B)** the computer that routes the data / जो computer data को route करता है
(C) the computer that terminates the data / जो computer data को terminate (समाप्त) करता है **(D)** all of the mentioned / उपर्युक्त सभी (all of the above)

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q30] · Hard

Q18. Consider the statements: S1: The physical medium of connection is known as a link. S2: The computers connected to it are known as nodes.

कथनों पर विचार करें: S1: connection का physical medium link कहलाता है। S2: इससे जुड़े computers nodes कहलाते हैं।

(A) Both S1 and S2 are false / दोनों S1 और S2 ग़लत हैं **(B)** S1 true but S2 false / S1 सही है लेकिन S2 ग़लत है
(C) S1 false but S2 true / S1 ग़लत है लेकिन S2 सही है **(D)** Both S1 and S2 true / दोनों S1 और S2 सही हैं

[Source: Repo: CN Topic Test-02 (UNIT TEST-2), Q2] · Hard

Q19. Which one of the following is NOT one of the five basic components of a data communication system?

निम्नलिखित में से कौन data communication system के five basic components में से एक नहीं है?

(A) Message / Message (message) **(B)** Protocol / Protocol (protocol)
(C) Router / Router (router) **(D)** Transmission medium / Transmission medium (medium)

[Source: Extra (Forouzan) - fills syllabus gap] · Hard

Q20. Which of the following is NOT a component of data communication?

निम्न में से कौन data communication का component नहीं है?

(A) Message **(B)** Protocol
(C) Compiler **(D)** Transmission medium

[Source: Practice — original (textbook-based)] · Hard

Section B - Characteristics (Delivery, Accuracy, Timeliness, Jitter) & Criteria (Performance, Reliability, Security)

Q21. Which physical layer impairment mainly causes jitter in VoIP traffic?

VoIP traffic में jitter का मुख्य kaaran कौन सा physical layer impairment है?

(A) Attenuation / attenuation (signal की shakti कम होना)
(B) Delay variation / delay का अंतर (variation)
(C) Distortion / distortion (vikriti) **(D)** Crosstalk

[Source: Repo: BCI Subject Test-8 (UNIT TEST-3 BCI/SCI), Q36] · Easy

Q22. What does jitter refer to in data communication?

Data communication में jitter का मतलब क्या होता है?

(A) Loss of data / data का loss **(B)** Delay variation in packets / packets में delay का अंतर (variation)
(C) Data synchronization / data का synchronization **(D)** Data encoding / data का encoding

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q1] · Medium

Q23. Which of the following is a characteristic of data communication?

निम्नलिखित में से कौन data communication की एक characteristic (विशेषता) है?

(A) Timeliness / समयपालन (timeliness) **(B)** Delay / دیر (delay)
(C) Speed / गति (speed) **(D)** Size / आकार (size)

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q6] · Medium

Q24. What is the maximum data rate of a transmission medium known as?

Transmission medium की अधिकतम data rate को क्या कहा जाता है?

(A) Bandwidth **(B)** Throughput
(C) Latency **(D)** Jitter

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q9] · Medium

Q25. Speed of an internet connection is measured in:

Internet connection की speed किसमें मापी जाती है?

(A) Kilobits per second (kbps) **(B)** Kilobytes per second (KB/s) only / केवल Kilobytes per second (KB/s)
(C) Metres per second **(D)** Hertz

[Source: Repo: Unit Test-4 Solution (OCR), p19] · Medium

Q26. Frequency of failure and network recovery time after a failure are measures of the _____ of a network.

Failure की आवृत्ति (frequency) और failure के बाद network recovery time किस चीज़ के measures हैं?

(A) Performance **(B)** Reliability
(C) Security **(D)** Feasibility

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q39] · Medium

Q27. What is the criteria to check the reliability of a network? 1. Downtime 2. Failure Frequency 3. Fault Tolerance 4. Security

Network की reliability check करने के criteria क्या हैं? 1. Downtime 2. Failure Frequency 3. Fault Tolerance 4. Security

(A) 1, 2, 3 only / सिर्फ 1, 2, 3 **(B)** 2, 3, 4 only / सिर्फ 2, 3, 4
(C) 1, 2, 4 only / सिर्फ 1, 2, 4 **(D)** 1, 2, 3, 4 only / 1, 2, 3, 4 सभी

[Source: Repo: CN Topic Test-02 (UNIT TEST-2), Q1] · Medium

Q28. What are the three criteria necessary for an effective and efficient network?

Effective और efficient network के लिए आवश्यक तीन criteria क्या हैं?

(A) Performance, Reliability, Security **(B)** Delivery, Accuracy, Timeliness
(C) Speed, Cost, Size **(D)** Message, Sender, Receiver

[Source: Forouzan Ch1 (Q1-2) — Practice Set] · Medium

Q29. What network property is achieved through the use of the Automatic Repeat reQuest (ARQ) mechanism?

Automatic Repeat reQuest (ARQ) mechanism से network की कौन सी property हासिल होती है?

(A) Reliability **(B)** Availability
(C) Confidentiality **(D)** Scalability

[Source: Repo: Random Test- Rajasthan PYQ (RPSC/RSSB)] · Medium

Q30. For effective data communication, the system must deliver data to the correct destination. This characteristic is called _____.

Effective data communication के लिए system को data सही destination पर पहुँचाना चाहिए। Is characteristic को _____ कहते हैं।

(A) Accuracy **(B)** Timeliness
(C) Delivery **(D)** Jitter

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

Q31. Data delivered late is useless (especially in real-time audio/video). Delivering data in a timely manner is called _____.

देर से पहुँचने वाला data बेकार है (ख़ास कर real-time audio/video में)। Time पर data pahunchane को _____ कहते हैं।

(A) Delivery **(B)** Accuracy
(C) Timeliness **(D)** Jitter

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

Q32. The network criterion measured by transit time, response time, throughput and delay is called _____.

Transit time, response time, throughput और delay से maape jane वाले network criterion को _____ कहते हैं।

(A) Performance **(B)** Reliability
(C) Security **(D)** Delivery

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

Q33. If packets arrive at the receiver with uneven delay, this problem is called:

यदि packets receiver पर असमान delay से पहुँचें, तो इस समस्या को क्या कहते हैं?

(A) Attenuation **(B)** Jitter
(C) Distortion **(D)** Noise

[Source: Practice — original (textbook-based)] · Medium

Q34. The amount of data that is actually delivered successfully per unit time is called:

प्रति इकाई समय में सफलतापूर्वक पहुँचा data कहलाता है:

(A) Bandwidth **(B)** Latency
(C) Throughput **(D)** Jitter

[Source: Practice — original (textbook-based)] · Medium

Q35. MTBF (Mean Time Between Failures) is related to which criteria of a network?

MTBF network के किस criteria से संबंधित है?

(A) Performance **(B)** Reliability
(C) Security **(D)** Accuracy

[Source: Practice — original (textbook-based)] · Medium

Q36. Accuracy in data communication means:

Data communication में accuracy का अर्थ है:

(A) Data is sent quickly / data तेज़ी से भेजा जाता है **(B)** Data is not corrupted / data corrupted (खराब) नहीं होता
(C) Data is encrypted / data encrypt होता है **(D)** Data reaches the receiver / data receiver तक pahunchta है

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q8] · Hard

Q37. An unauthorized user is a network _____ issue.

An unauthorized user एक network _____ issue है.

(A) Performance **(B)** Reliability
(C) Security **(D)** All of the above / ऊपर दिए गए सभी

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q40] · Hard

Q38. Which of the following is NOT one of the three fundamental criteria (properties) that a network must meet?

निम्नलिखित में से कौन network के तीन fundamental criteria (properties) में से एक नहीं है?

(A) Performance **(B)** Reliability
(C) Security **(D)** Topology

[Source: Extra (Forouzan) - fills syllabus gap] · Hard

Q39. The capacity of a communication channel is measured in:

Communication channel की क्षमता मापी जाती है:

(A) Bandwidth **(B)** Bit capacity
(C) Volume **(D)** Noise

[Source: KVS PGT/TGT/PRT computer banks 2018 (via jagranjosh)] · Hard

Q40. Consider a 100 Mbps link between an earth station (sender) and a satellite (receiver) at an altitude of 2100 km. The signal propagates at 3×10^8 m/s. The time taken (in milliseconds, rounded to two decimal places) for the receiver to completely receive a 1000-byte packet transmitted by the sender is _____.

100 Mbps के link पर earth station से 2100 km ऊँचे satellite को 1000-byte packet भेजा गया; signal की speed 3×10^8 m/s है। Receiver को packet पूरा मिलने में लगा समय (ms में, दो दशमलव) कितना है?

(A) 7.08 **(B)** 2.10
(C) 7.00 **(D)** 21.08

[Source: GATE CS 2022] · Hard

Q41. What is the bit rate for transmitting an uncompressed 800×600 pixel colour frame with 8 bits/pixel at 40 frames per second?

बिना compression के 800×600 pixel वाला colour frame (8 bits/pixel) 40 frames per second पर transmit हो — bit rate कितना होगा?

(A) 2.4 Mbps **(B)** 153.6 Mbps
(C) 15.36 Mbps **(D)** 1536 Mbps

[Source: KVS PGT Computer Science 2017] · Hard

Q42. An analog signal carries 4 bits in each signal unit. If 1000 signal units are sent per second, then the baud rate and bit rate of the signal are _____ and _____.

एक analog signal हर signal unit में 4 bits रखता है। 1000 signal units per second भेजे जाएँ तो signal की baud rate और bit rate क्रमशः कितनी होगी?

(A) 4000 bauds/sec, 4000 bps **(B)** 1000 bauds/sec, 1000 bps
(C) 4000 bauds/sec, 1000 bps **(D)** 1000 bauds/sec, 4000 bps

[Source: KVS PGT Computer Science 2017] · Hard

Section C - Transmission Mode (Direction of Data Flow: Simplex, Half-Duplex, Full-Duplex)

Q43. Communication between a computer and a keyboard involves _____ transmission.

Computer और keyboard के बीच communication में _____ transmission होता है।

(A) Automatic **(B)** Half-duplex
(C) Half-duplex (variant) / half-duplex **(D)** Simplex

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q33] · Medium

Q44. A television broadcast is an example of _____ transmission.

Television broadcast _____ transmission का एक example है।

(A) simplex **(B)** half-duplex
(C) full-duplex **(D)** automatic

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q41] · Medium

Q45. What is the mode of transmission where data flows in only one direction?

Wo transmission mode kaunsa है जिसमें data सिर्फ एक direction में flow करता है?

(A) Simplex **(B)** Half-Duplex
(C) Full-Duplex **(D)** Multiplex

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q3] · Medium

Q46. Which transmission mode is used in walkie-talkie communication?

Walkie-talkie communication में कौन सा transmission mode use होता है?

(A) Simplex **(B)** Half-Duplex
(C) Full-Duplex **(D)** Multiplex

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q10] · Medium

Q47. Which of the following statements are TRUE? 1. Hubs operate in Half-Duplex mode. 2. Switches operate in Full-Duplex mode. 3. Routers broadcast by default.

निम्नलिखित में से कौन से statements TRUE हैं? 1. Hubs Half-Duplex mode में काम करते हैं। 2. Switches Full-Duplex mode में काम करते हैं। 3. Routers by default broadcast करते हैं।

(A) 1 and 2 / 1 और 2 **(B)** 2 and 3 / 2 और 3
(C) 1 and 3 / 1 और 3 **(D)** 1, 2 and 3 / 1, 2 और 3

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Medium

Q48. Which of the following works in simplex mode?

निम्न में से कौन simplex mode में काम करता है?

(A) Telephone **(B)** Walkie-talkie
(C) Keyboard and monitor / Keyboard और monitor **(D)** Chat application

[Source: Practice — original (textbook-based)] · Medium

Q49. Which of the following transmission directions listed is not a legitimate channel?

निम्नलिखित में से कौन सा transmission direction एक legitimate (vaidh) channel नहीं है?

(A) Simplex **(B)** Half Duplex
(C) Full Duplex **(D)** Double Duplex

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q3] · Hard

Q50. What is the difference between half-duplex and full-duplex transmission modes?

Half-Duplex और Full-Duplex transmission modes में अंतर क्या है?

(A) Half-duplex allows only one-way flow; full-duplex also allows only one-way flow / Half-Duplex में सिर्फ one-way flow; Full-Duplex में भी सिर्फ one-way flow **(B)** Half-duplex flows both ways one direction at a time; full-duplex flows both ways at the same time / Half-Duplex में data दोनों तरफ़ पर एक समय में एक direction; Full-Duplex में एक साथ दोनों तरफ़

(C) Half-duplex is only for wireless; full-duplex only for wired networks / Half-Duplex सिर्फ wireless के लिए; Full-Duplex सिर्फ wired के लिए **(D)** There is no difference between the two / दोनों में कोई अंतर नहीं है

[Source: Forouzan Ch1 (Q1-6) — Practice Set] · Hard

Q51. Data flow between two devices can occur in a _____ way.

Do devices के बीच data flow _____ tareeke से हो सकता है।

(A) simplex **(B)** half-duplex
(C) full-duplex **(D)** all of the above / ऊपर दिए गए सभी

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q45] · Hard

Q52. In a telephone conversation, both persons can speak and hear each other at the same time. This is an example of _____.

Telephone conversation में दोनों persons एक साथ bol और sun सकते हैं। यह _____ का example है।

(A) Simplex **(B)** Half-duplex
(C) Full-duplex **(D)** Unicast

[Source: Extra (Forouzan) - fills syllabus gap] · Hard

Q53. In which mode can both stations transmit and receive at the same instant?

किस mode में दोनों stations एक ही समय में भेज और पा सकते हैं?

(A) Simplex **(B)** Half-duplex
(C) Full-duplex **(D)** Broadcast

[Source: Practice — original (textbook-based)] · Hard

Section D - Network Structure / Topology (Physical: Mesh, Star, Bus, Ring, Others; Logical: IEEE 802)

Q54. IEEE 802.2 stअंदरd is related to which sub-layer?

IEEE 802.2 stअंदरd किस sub-layer से संबंधित है?

(A) MAC Sublayer **(B)** LLC Sublayer
(C) Physical Layer **(D)** Network Layer

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

Q55. What are the advantages of star topology?

Star topology के क्या advantages (लाभ) हैं?

(A) Less number of cables are required / कम number में cables चाहिए **(B)** More number of cables are required / ज़्यादा number में cables चाहिए
(C) It is a distributed topology / यह एक distributed topology है **(D)** Security is high / security ज़्यादा है

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q1] · Medium

Q56. Which type of topology is best suited for large businesses which must carefully control and coordinate the operation of distributed branch outlets?

Large businesses के लिए कौन सी topology सबसे उपयुक्त है, jahan distributed branch outlets के operation को carefully control और coordinate करना हो?

(A) Ring **(B)** Local area
(C) Hierarchical / hierarchical (tree) **(D)** Star

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q2] · Medium

Q57. The repeater is used in _____ topology.

Repeater का use _____ topology में होता है।

(A) Mesh Topology **(B)** Ring Topology
(C) Bus Topology **(D)** Star Topology

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q50] · Medium

Q58. What type of topology connects each node in a closed loop?

कौन सी topology हर node को एक closed loop में जोड़ती है?

(A) Star **(B)** Mesh
(C) Ring **(D)** Bus

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q4] · Medium

Q59. In a _____ topology, every device has a dedicated point-to-point link to every other device. For n devices, the number of links is n(n-1)/2 and each device needs n-1 I/O ports.

_____ topology में हर device का हर दूसरे device से dedicated point-to-point link होता है। n devices के लिए links की संख्या n(n-1)/2 होती है और हर device को n-1 I/O ports चाहिए।

(A) Bus **(B)** Star
(C) Ring **(D)** Mesh

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

Q60. In a _____ topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub (or switch).

_____ topology में हर device का सिर्फ एक central controller (hub/switch) से dedicated point-to-point link होता है।

(A) Mesh **(B)** Ring
(C) Star **(D)** Bus

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

Q61. Which network topology requires a multipoint connection?

किस network topology में multipoint connection की आवश्यकता होती है?

(A) Ring **(B)** Bus
(C) Star **(D)** Mesh

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

Q62. In a _____ topology, each device is connected to its two neighbours, data travels in one direction around the loop, and each device regenerates (repeats) the signal - this is called token passing.

_____ topology में हर device अपने दो neighbours से जुड़ा होता है, data loop में एक ही direction में जाता है, और हर device signal को regenerate (repeat) करता है - इसे token passing कहते हैं।

(A) Bus **(B)** Ring
(C) Star **(D)** Mesh

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

Q63. A network in which two or more different topologies are combined (e.g., a star connected to another star) is called a _____ topology, while a _____ topology is a hierarchy of stars (a variation of the star topology).

जिसमें do या ज़्यादा अलग topologies combine हो (जैसे एक star दूसरे star से जुड़ा हो), use _____ topology कहते हैं, और _____ topology stars की hierarchy है (star topology का variation).

(A) Hybrid ; Tree **(B)** Tree ; Hybrid
(C) Mesh ; Ring **(D)** Bus ; Star

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

Q64. Hierarchical topology, used to establish a network, is also known as a:

Hierarchical topology, जिससे network स्थापित किया जाता है, किस नाम से भी जानी जाती है?

(A) Bus network **(B)** Star network
(C) Tree network **(D)** Mesh network

[Source: Repo: Unit Test-4 Solution (OCR), p17] · Medium

Q65. The _____ topology of a network defines how the devices are physically laid out/ connected, whereas the _____ topology defines how the data actually flow / how the devices access the network to transmit.

Network की _____ topology बताती है की devices physically कैसे जुड़े हैं, jabki _____ topology बताती है की data asal में कैसे flow करता है / devices transmit करने के लिए network को कैसे access करते हैं।

(A) Logical ; Physical **(B)** Physical ; Logical
(C) Star ; Bus **(D)** Mesh ; Ring

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

Q66. Which physical topology offers the highest robustness (fault tolerance) because of multiple redundant paths, but is also the most expensive and hardest to install because of the huge amount of cabling?

कौन सी physical topology सबसे ज़्यादा robust (fault tolerant) है क्योंकि ismein कई redundant paths होते हैं, लेकिन सबसे महंगी और install करने में सबसे मुश्किल भी है क्योंकि बहुत ज़्यादा cabling चाहिए?

(A) Bus **(B)** Ring
(C) Star **(D)** Mesh

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

Q67. If the central device of a star topology fails, the result is:

यदि star topology का central device ख़राब हो जाए तो:

(A) Only one device stops / केवल एक device रुकता है **(B)** Whole network stops / पूरा network रुक जाता है
(C) Network works as mesh / Network mesh जैसा चलता है
(D) Only data transfer slows / केवल data transfer धीमा होता है

[Source: Practice — original (textbook-based)] · Medium

Q68. Which topology gives every device equal access using a token?

कौन-सी topology token द्वारा हर device को बराबर पहुँच देती है?

(A) Bus **(B)** Star
(C) Ring **(D)** Mesh

[Source: Practice — original (textbook-based)] · Medium

Q69. A hybrid topology is:

Hybrid topology है:

(A) Only mesh / केवल mesh **(B)** Combination of two or more topologies / दो या अधिक topologies का मिश्रण
(C) Only bus / केवल bus **(D)** A wireless topology / एक wireless topology

[Source: Practice — original (textbook-based)] · Medium

Q70. Which one of the following is not a network topology?

निम्नलिखित में से कौन एक network topology नहीं है?

(A) Star **(B)** Ring
(C) Bus **(D)** Peer to Peer

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q9] · Hard

Q71. For n devices in a network, how many cable links are required in a mesh topology?

Network में n devices के लिए mesh topology में कितने cable links चाहिए?

(A) n **(B)** n - 1
(C) n(n - 1)/2 **(D)** n(n - 1)

[Source: Forouzan Ch1 (Q1-8) — Practice Set] · Hard

Q72. Six devices are arranged in a mesh topology. How many cables are needed, and how many ports does each device need?

छह devices को mesh topology में जोड़ा गया है। कितनी cables चाहिए और हर device को कितने ports चाहिए?

(A) 15 cables, 5 ports each / 15 cables, हर device पर 5 ports
(B) 30 cables, 6 ports each / 30 cables, हर device पर 6 ports
(C) 15 cables, 6 ports each / 15 cables, हर device पर 6 ports
(D) 6 cables, 5 ports each / 6 cables, हर device पर 5 ports

[Source: Forouzan Ch1 (P1-3) — Practice Set (Problem)] · Hard

Q73. If a computer network cable breaks, the whole network goes down in which topology?

यदि computer network cable टूट जाए, तो किस topology में पूरा network बंद हो जाता है?

(A) Star **(B)** Ring
(C) Mesh **(D)** All of these / ये सभी

[Source: Repo: Unit Test-4 Solution (OCR), p16] · Hard

Q74. In which topology does the failure of a single cable or device NOT affect the rest of the network?

किस topology में एक cable या device के fail होने पर बाकी network प्रभावित नहीं होता?

(A) Bus **(B)** Ring
(C) Star **(D)** Mesh

[Source: Forouzan Ch1 (P1-4) — Practice Set (Problem)] · Hard

Q75. In a _____ topology, one long cable acts as a backbone to link all the devices; devices connect via drop lines and taps, and terminators are needed at both ends of the backbone.

_____ topology में एक लंबा cable backbone की तरह सभी devices को जोड़ता है; devices drop lines और taps से connect होते हैं, और backbone के दोनों ends पर terminators चाहिए।

(A) Bus **(B)** Star
(C) Ring **(D)** Mesh

[Source: Extra (Forouzan) - fills syllabus gap] · Hard

Q76. The IEEE 802 stअंदरds (such as 802.3 for Ethernet, 802.5 for Token Ring, 802.11 for Wi-Fi) primarily define the _____ of LANs/MANs.

IEEE 802 stअंदरds (जैसे 802.3 Ethernet, 802.5 Token Ring, 802.11 Wi-Fi) primarily LANs/MANs की _____ define करते हैं।

(A) Physical and Data Link layers (including the logical topology / MAC) / Physical और Data Link layers (logical topology / MAC sahit) **(B)** Only the Application layer / सिर्फ Application layer
(C) Only IP addressing / सिर्फ IP addressing **(D)** Only WAN routing / सिर्फ WAN routing

[Source: Extra (Forouzan) - fills syllabus gap] · Hard

Q77. Four network segments in Bus topology are: A1–A2, B1–B2–B3, C1–C2 and D1–D2–D3. If A1, B1, C1 and D1 are interconnected in a mesh, how many mesh connections are required?

Bus topology में चार network segments हैं: A1–A2, B1–B2–B3, C1–C2 और D1–D2–D3। यदि A1, B1, C1 और D1 को mesh में जोड़ा जाए, तो कितने mesh connections चाहिए?

(A) 12 **(B)** 10
(C) 6 **(D)** 45

[Source: RPSC Programmer 2024 Paper-1 (with official key)] · Hard

Q78. In a mesh topology with 6 devices, how many dedicated links are required?

6 devices की mesh topology में कितने dedicated links चाहिए?

(A) 15 **(B)** 30
(C) 36 **(D)** 12

[Source: Practice — original (textbook-based)] · Hard

Section E - Types of Networks (LAN, MAN, WAN, PAN, CAN, SAN; Internet, Intranet, Extranet; Point-to-Point, Multipoint; Client-Server, P2P)

Q79. The term WAN stands for?

WAN का पूरा नाम (full form) क्या है?

(A) Wide Area Net **(B)** Wide Access Network
(C) Wide Area Network **(D)** Wide Access Net

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q11 / BCI_SCI Subject Test-01, Q11] · Medium

Q80. Which network is suitable for a building?

किसी एक building के लिए कौन सा network उपयुक्त (suitable) है?

(A) WAN **(B)** LAN
(C) MAN **(D)** PAN

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q27] · Medium

Q81. Bluetooth is an example of:

Bluetooth इसका example है:

(A) personal area network / personal area network (PAN)
(B) local area network / local area network (LAN)
(C) virtual private network / virtual private network (VPN)
(D) none of the mentioned / इनमें से कोई नहीं

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q31] · Medium

Q82. Which one of the following extends a private network across public networks?

निम्नलिखित में से कौन एक private network को public networks के paar (across) extend करता है?

(A) local area network / local area network (LAN) **(B)** virtual private network / virtual private network (VPN)
(C) enterprise private network **(D)** storage area network / storage area network (SAN)

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q32] · Medium

Q83. _____ is a network that covers geographic areas that are larger, such as districts or cities.

_____ एक ऐसा network है जो बड़े geographical areas, जैसे districts या cities, को cover करता है।

(A) LAN **(B)** MAN
(C) WAN **(D)** None of these / इनमें से कोई नहीं

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q49] · Medium

Q84. Which factor primarily decides whether a communication system is a LAN or a WAN?

कौन-सा factor मुख्य रूप से तय करता है कि communication system LAN है या WAN?

(A) The geographical area it spans / वह कितने geographical area में फैला है **(B)** The number of protocols it uses / वह कितने protocols use करता है
(C) The colour of the cables / Cables का रंग **(D)** The type of keyboard attached / जुड़े keyboard का प्रकार

[Source: Forouzan Ch1 (Q1-9) — Practice Set] · Medium

Q85. In terms of the size of the network, the correct order (ascending) is:

Network के size के hisaab से सही kram (badhta हुआ/ ascending order) क्या है?

(A) PAN, MAN, LAN, WAN **(B)** LAN, MAN, WAN, PAN
(C) PAN, LAN, MAN, WAN **(D)** LAN, PAN, MAN, WAN

[Source: Repo: CN Topic Test-02 (UNIT TEST-2), Q42] · Medium

Q86. Which type of network covers an entire city or campus?

कौन सा network पूरे एक city या campus को cover करता है?

(A) LAN **(B)** MAN
(C) WAN **(D)** PAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q2] · Medium

Q87. The Internet is an example of:

Internet इसका example है:

(A) LAN **(B)** MAN
(C) WAN **(D)** PAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q12] · Medium

Q88. SAN is mainly used for:

SAN का मुख्य उपयोग किसके लिए होता है?

(A) File Sharing **(B)** Storage Management
(C) Internet Access **(D)** Printing Documents / documents print करना

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q13] · Medium

Q89. PAN is used for connecting:

PAN किसके connect करने के लिए use होता है?

(A) Personal devices like mobile, laptop, Bluetooth headset / personal devices जैसे mobile, laptop, Bluetooth headset **(B)** Computers in a company / एक company के computers

(C) Computers in a city / एक city के computers **(D)** Global offices

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q14] · Medium

Q90. A school has four computer labs in the same building. Each lab has 20 computers connected to a central server. Students can share files and print from any lab. Which type of network is implemented in this school?

Ek school में एक ही building में चार computer labs हैं. हर lab में 20 computers एक central server से जुड़े हैं। Students किसी भी lab से files share और print कर सकते हैं। School में कौन सा network implemented है?

(A) WAN **(B)** MAN

(C) LAN **(D)** CAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q15] · Medium

Q91. A university has different departments in separate buildings like Engineering, Arts, and Commerce. Each department has its own LAN. All these LANs are connected together for communication. Which type of network connects all these LANs?

Ek university में अलग-अलग departments अलग buildings में हैं जैसे Engineering, Arts, Commerce। हर department का apna LAN है। यह सब LANs communication के लिए आपस में जुड़े हैं। कौन सा network in सब LANs को जोड़ता है?

(A) LAN **(B)** CAN

(C) MAN **(D)** WAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q16] · Medium

Q92. An international bank connects its main office in New York with branches in Delhi, London, and Tokyo. Which network type is used here?

Ek international bank अपने main office (New York) को Delhi, London, और Tokyo की branches से जोड़ता है। यहाँ कौन सा network type use होता है?

(A) MAN **(B)** WAN

(C) CAN **(D)** LAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q17] · Medium

Q93. A company has several servers that store and manage large volumes of data. All storage devices are interconnected for high-speed data access. Which type of network is this?

Ek company के पास कई servers हैं जो बड़ी matra में data store और manage करते हैं। सब storage devices high-speed data access के लिए interconnected हैं। यह कौन सा network है?

(A) LAN **(B)** SAN

(C) CAN **(D)** WAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q18] · Medium

Q94. A supplier accessing a company's order tracking system uses:

Ek supplier जो company के order tracking system को access करता है, कौन सा network use करता है?

(A) Internet **(B)** Intranet

(C) Extranet **(D)** VPN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q19] · Medium

Q95. Which network type provides maximum security and privacy?

कौन सा network type अधिकतम (maximum) security और privacy provide करता है?

(A) Internet **(B)** Intranet

(C) Extranet **(D)** None

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q20] · Medium

Q96. A software company shares project reports and progress updates with its clients through a secure online portal that requires login credentials. Which network type is this?

Ek software company अपने clients के साथ project reports और progress updates एक secure online portal के through share करती है जिसमें login credentials चाहिए। यह कौन सा network type है?

(A) Internet **(B)** Intranet

(C) Extranet **(D)** MAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q21] · Medium

Q97. Three or more devices share a link in _____ connection.

Teen या अधिक devices एक link _____ connection में share करते हैं।

(A) Unipoint **(B)** Multipoint
(C) Point to point **(D)** None of the mentioned / इनमें से कोई नहीं

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q36] · Medium

Q98. What are the two types of line configuration?

Line configuration के दो प्रकार क्या हैं?

(A) Simplex and Half-Duplex / Simplex और Half-Duplex
(B) Serial and Parallel / Serial और Parallel
(C) Point-to-Point and Multipoint / Point-to-Point और Multipoint **(D)** Analog and Digital / Analog और Digital

[Source: Forouzan Ch1 (Q1-4) — Practice Set] · Medium

Q99. What is the advantage of a multipoint connection over a point-to-point connection?

Point-to-point connection की तुलना में multipoint connection का क्या लाभ है?

(A) Fewer cables are needed, so the cost is lower / कम cables लगती हैं, इसलिए cost कम होती है **(B)** It provides a dedicated link for every pair of devices / हर device-युग्म को dedicated link मिलता है
(C) It is more secure than point-to-point / यह point-to-point से ज़्यादा secure है **(D)** It needs more cables and ports / इसमें ज़्यादा cables और ports लगते हैं

[Source: Forouzan Ch1 (Q1-3) — Practice Set] · Medium

Q100. A _____ connection provides a dedicated link between two devices.

Ek _____ connection do devices के बीच एक dedicated link provide करता है।

(A) point-to-point **(B)** multipoint
(C) primary **(D)** secondary

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q42] · Medium

Q101. When a person makes a local telephone call to another person, what type of connection is it?

जब कोई व्यक्ति दूसरे व्यक्ति को local telephone call करता है, तो यह किस प्रकार का connection है?

(A) Multipoint connection **(B)** Point-to-point connection
(C) Broadcast connection **(D)** Multicast connection

[Source: Forouzan Ch1 (P1-7) — Practice Set (Problem)] · Medium

Q102. Which of the following is an example of Client-Server architecture?

निम्नलिखित में से कौन Client-Server architecture का example है?

(A) Chat between two Bluetooth devices / do Bluetooth devices के बीच chat **(B)** Web browser accessing a website / web browser जो एक website access कर रहा है
(C) File transfer via USB cable / USB cable से file transfer
(D) Two laptops connected directly / do laptops directly connected

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q23] · Medium

Q103. In a bank, customers use ATMs to check balance or withdraw cash. ATM machines send the request to a central computer for verification and processing. Which type of architecture is used here?

Ek bank में customers ATMs से balance check या cash withdraw करते हैं। ATM machines request एक central computer को verification और processing के लिए bhejte हैं। यहाँ कौन सी architecture use होती है?

(A) Peer-to-Peer **(B)** Client-Server
(C) Multipoint **(D)** Point-to-Point

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q25] · Medium

Q104. In a LAN, connectivity can be created using Wi-Fi or _____.

Ek LAN में connectivity Wi-Fi या _____ से बनाई जा सकती है।

(A) TV **(B)** Radio
(C) Mobile **(D)** Ethernet

[Source: Repo: Random Test- Rajasthan PYQ (RPSC/RSSB)] · Medium

Q105. Which network type is built on top of another existing network?

कौन-सा network type किसी मौजूदा network के ऊपर बनाया जाता है?

(A) Overlay network **(B)** Prime network
(C) Prior network **(D)** Chief network

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

Q106. _____ converts analog voice signals into digital and then transmits them over the broadband line.

_____ analog voice signals को digital में बदलकर broadband line पर भेजता है.

(A) VoIP (Voice over Internet Protocol) **(B)** Chatting
(C) Website **(D)** Video conferencing

[Source: RSMSSB Informatics Assistant 2023 (via examsmantra solved paper)] · Medium

Q107. A network that connects storage devices and servers in a data centre is called:

Data centre में storage devices और servers को जोड़ने वाला network है:

(A) PAN **(B)** CAN
(C) SAN **(D)** MAN

[Source: Practice — original (textbook-based)] · Medium

Q108. A company's internal network that uses Internet technology but is accessible only to its employees is called:

Internet technology पर चलने वाला वह internal network जो केवल कर्मचारियों के लिए हो:

(A) Internet **(B)** Extranet
(C) Intranet **(D)** WAN

[Source: Practice — original (textbook-based)] · Medium

Q109. In a peer-to-peer network:

Peer-to-peer network में:

(A) One computer is always the server / एक computer हमेशा server होता है **(B)** Every computer can act as client and server / हर computer client और server दोनों बन सकता है **(C)** Only files can be shared / केवल files साझा होती हैं **(D)** It cannot work without a hub / Hub के बिना नहीं चलता

[Source: Practice — original (textbook-based)] · Medium

Q110. When several senders share one communication line through a common path, the access is called:

जब कई senders एक ही line साझा करते हैं, तो वह access है:

(A) Point-to-point **(B)** Dedicated
(C) Multipoint **(D)** Simplex

[Source: Practice — original (textbook-based)] · Medium

Q111. Which of these is NOT a type of network?

इनमें से कौन network का एक prakar (type) नहीं है?

(A) LAN **(B)** MAN
(C) WAN **(D)** TAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q7] · Hard

Q112. What is the difference between an 'internet' and the 'Internet'?

'internet' और 'Internet' में अंतर क्या है?

(A) There is no difference; both mean the World Wide Web / कोई अंतर नहीं; दोनों का अर्थ World Wide Web है **(B)** An internet is any network of networks; the Internet is the specific global one we use today / internet कोई भी network of networks है; Internet आज की वह विशिष्ट विश्वव्यापी network of networks है

(C) An internet is smaller than a LAN; the Internet is larger than a WAN / internet LAN से छोटा है; Internet WAN से बड़ा है **(D)** An internet uses cables; the Internet uses only wireless / internet cables use करता है; Internet सिर्फ wireless use करता है

[Source: Forouzan Ch1 (Q1-10) — Practice Set] · Hard

Q113. Bank employees access internal financial databases, while customers access their accounts through online banking websites. Which networks are being used by employees and customers respectively?

Bank employees internal financial databases access करते हैं, jabki customers online banking websites के through अपने accounts access करते हैं। Employees और customers kramashah कौन से networks use कर रहे हैं?

(A) Employees - Intranet; Customers - Extranet **(B)** Employees - Internet; Customers - Intranet
(C) Employees - Extranet; Customers - Internet **(D)** Both - Internet / दोनों - Internet

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q22] · Hard

Q114. How many point-to-point WANs are needed to connect n LANs so that every LAN can directly communicate with every other LAN?

n LANs को इस तरह जोड़ने के लिए कितने point-to-point WANs चाहिए कि हर LAN हर दूसरी LAN से सीधे communicate कर सके?

(A) n **(B)** n - 1
(C) n(n - 1)/2 **(D)** n^2

[Source: Forouzan Ch1 (Q1-13) — Practice Set] · Hard

Q115. Which of the following uses Peer-to-Peer (P2P) architecture, not Client-Server?

निम्नलिखित में से कौन Peer-to-Peer (P2P) architecture use करता है, Client-Server नहीं?

(A) BitTorrent file sharing **(B)** Google search
(C) Gmail **(D)** YouTube

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q24] · Hard

Section F - Network History (ARPANET to Today's Network)

Q116. ARPANET stands for _____.

ARPANET का पूरा नाम है _____.

(A) Advanced Recheck Projects Agency Internet **(B)** Advanced Recheck Projects Agency Network
(C) Advanced Research Projects Agency Network **(D)** Advanced Research Projects Agency Internet

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q26] · Medium

Q117. The first network was:

सबसे पहला network था:

(A) CNNET **(B)** NSFNET
(C) ASAPNET **(D)** ARPANET

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q34] · Medium

Q118. In the original ARPANET, _____ were directly connected together.

Original ARPANET में, _____ आपस में directly connected the।

(A) IMPs / IMPs (Interface Message Processors) **(B)** host computers
(C) networks **(D)** routers

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q43] · Medium

Q119. The first operational computer network in the world was the _____ for the United States Department of Defense.

दुनिया का पहला operational computer network, United States Department of Defense के लिए _____ था।

(A) ARPANET **(B)** ERNET
(C) SKYNET **(D)** DARPANET

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q48] · Medium

Q120. ARPANET - the precursor of today's Internet - was created in _____ by ARPA (Advanced Research Projects Agency) of the U.S. Department of Defense, starting with four nodes (UCLA, SRI, UC Santa Barbara, University of Utah).

ARPANET - aaj के Internet का purvaj - _____ में ARPA (Advanced Research Projects Agency), U.S. Department of Defense द्वारा बनाया गया था, चार nodes से shuruaat हुई (UCLA, SRI, UC Santa Barbara, University of Utah).

(A) 1957 **(B)** 1969
(C) 1983 **(D)** 1991

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

Q121. TCP/IP (developed by Vint Cerf and Bob Kahn) was adopted by ARPANET as its official protocol suite on 1 January _____ - an event known as the fflag day.

TCP/IP (Vint Cerf और Bob Kahn द्वारा develop किया गया) को ARPANET ने apna official protocol suite बनाया था 1 January _____ को - is event को fflag day कहा जाता है।

(A) 1969 **(B)** 1972
(C) 1983 **(D)** 1995

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

Q122. NSFNET, created by the U.S. National Science Foundation in _____, became a major backbone of the Internet, connecting universities and research centres.

NSFNET, U.S. National Science Foundation द्वारा _____ में बनाया गया, Internet का एक major backbone बन गया, universities और research centres को जोड़ता था।

(A) 1969 **(B)** 1983
(C) 1986 **(D)** 1991

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

Q123. The World Wide Web (WWW) was invented in 1991 at CERN by _____, and it made the Internet popular among the general public.

World Wide Web (WWW) का invention 1991 में CERN में _____ ने किया था, और isne Internet को आम public के बीच popular बनाया।

(A) Vint Cerf **(B)** Tim Berners-Lee
(C) Bob Kahn **(D)** Marc Andreessen

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

Q124. ARPANET, the ancestor of today's Internet, became operational in:

आज के Internet का पूर्वज ARPANET किस वर्ष चालू हुआ?

(A) 1969 **(B)** 1983
(C) 1991 **(D)** 1975

[Source: Practice — original (textbook-based)] · Medium

Q125. The protocol suite used by today's Internet was developed as part of:

आज के Internet का protocol suite किस परियोजना का हिस्सा था?

(A) NSFNET **(B)** ARPANET/TCP-IP project / ARPANET/TCP-IP परियोजना
(C) Ethernet project / Ethernet परियोजना **(D)** ISO project / ISO परियोजना

[Source: Practice — original (textbook-based)] · Medium

Section G - General Network Basics (WWW, E-mail, Protocols, Servers, Switching - from repo intro tests)

Q126. A collection of hyperlinked documents on the internet forms the?

Internet पर hyperlinked documents का sangrah (collection) क्या banata है?

(A) World Wide Web (WWW) **(B)** E-mail system
(C) Mailing list **(D)** Hypertext markup language / hypertext markup language (HTML)

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q5] · Medium

Q127. Which one of the following is a valid email address?

निम्नलिखित में से कौन सा एक valid email address है?

(A) computech@academy.com **(B)** gmail.com
(C) compu@.com **(D)** computech@books

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q6] · Medium

Q128. Which of the following best describes uploading information?

निम्नलिखित में से कौन uploading information को सबसे achhe से describe करता है?

(A) Sorting data on a disk drive / disk drive पर data को sort करना **(B)** Sending information to a host computer / information को एक host computer को भेजना
(C) Receiving information from a host computer / एक host computer से information receive करना **(D)** Sorting data on a hard drive / hard drive पर data को sort करना

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q7] · Medium

Q129. Which one of the following is the most common internet protocol?

Nimnlikhis में से कौन सा सबसे common internet protocol है?

(A) HTML **(B)** NetBEUI
(C) TCP/IP **(D)** IPX/SPX

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q8] · Medium

Q130. The dial-up connections to access the internet use ___ protocol.

Internet access के लिए dial-up connections ___ protocol use करते हैं।

(A) PPP (Point to Point protocol) **(B)** X.25 protocol
(C) PLIP protocol **(D)** None of the above / इनमें से कोई नहीं

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q18] · Medium

Q131. When a resident uses a dial-up or DSL service to connect to the Internet, what is the role of the telephone company?

जब कोई व्यक्ति Internet से जुड़ने के लिए dial-up या DSL service use करता है, तो telephone company की भूमिका क्या होती है?

(A) It provides the switching/connection path from the user to the ISP / वह user से ISP तक switching/connection path देती है **(B)** It hosts every website the user visits / वह user की हर website host करती है
(C) It assigns IP addresses to every website / वह हर website को IP address देती है **(D)** It encrypts all the user's data / वह user का सारा data encrypt करती है

[Source: Forouzan Ch1 (Q1-14) — Practice Set] · Medium

Q132. Which of the following servers allows LAN users to share data?

निम्नलिखित में से कौन सा server LAN users को data share करने की अनुमति देता है?

(A) Data server **(B)** Point server
(C) File server **(D)** Communication server

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q22] · Medium

Q133. EDI stands for ____.

EDI का पूरा नाम है ____.

(A) Electronic Data Interchange **(B)** Electronic Digital Internet
(C) Electronic Digital Interchange **(D)** Electronic Data Internet

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q25] · Medium

Q134. The term UPnP is a short form of:

UPnP term का पूरा नाम (full form) क्या है?

(A) Uniform Plug and Play **(B)** Unit Plug and Play
(C) Useful Plug and Play **(D)** Universal Plug and Play

[Source: Repo: Unit Test-4 Solution (OCR), p11] · Medium

Q135. Consider the switching statements: 1. Circuit Switching establishes a dedicated path. 2. Packet Switching routes data in packets on different paths. 3. Message Switching uses store-and-forward. Which statements are correct?

Switching statements देखिए: 1. Circuit Switching dedicated path बनाता है। 2. Packet Switching data को packets में अलग paths पर भेजता है। 3. Message Switching store-and-forward use करता है। कौन से statements सही हैं?

(A) 1 and 2 only / सिर्फ 1 और 2 **(B)** 2 and 3 only / सिर्फ 2 और 3
(C) 1 and 3 only / सिर्फ 1 और 3 **(D)** 1, 2 and 3 / 1, 2 और 3

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Medium

Q136. All the information on the internet is passed using _____ switching method.

Internet पर saari information _____ switching method से pass होती है।

(A) Portable **(B)** Packets
(C) Pouch **(D)** Primary

[Source: Repo: Random Test- Rajasthan PYQ (RPSC/RSSB)] · Medium

Q137. Which technique breaks data into packets for transmission?

Transmission के लिए data को packets में तोड़ने की technique कौन-सी है?

(A) Circuit switching **(B)** Packet switching
(C) Frequency division multiplexing **(D)** Time division multiplexing

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)] · Medium

Q138. In telephone networks, a dedicated end-to-end path is established before communication. This is:

Telephone network में संचार से पहले समर्पित मार्ग बनता है। यह है:

(A) Packet switching **(B)** Circuit switching
(C) Message switching **(D)** Datagram switching

[Source: Practice — original (textbook-based)] · Medium

Q139. Home broadband connections that work on DSL (Digital Subscriber Line) use ___ protocol.

DSL (Digital Subscriber Line) पर काम करने वाले home broadband connections ___ protocol use करते हैं।

(A) PPPoE (PPP over Ethernet) **(B)** PPPoA (PPP over ATM)
(C) Bridge protocol **(D)** All of the above / ऊपर दिए गए सभी

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q19] · Hard

Q140. A computer network permits sharing of:

एक computer network किस चीज़ की sharing की अनुमति देता है?

(A) Resources / resources (printer, disk, etc.) **(B)** Information / information (data)
(C) Both a & b / दोनों a और b **(D)** None of these / इनमें से कोई नहीं

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q47 / CN Topic Test-02 (UNIT TEST-2), Q47] · Hard

Q141. Which of the following is a benefit of networking?

Networking का कौन-सा लाभ (benefit) है?

(A) File sharing **(B)** Easier access to resources / resources तक आसान पहुँच
(C) Easier backups / आसान backups **(D)** All of the above / ऊपर दिए गए सभी

[Source: InstrumentationTools 150 CN MCQs] · Hard

Q142. How many digits of the Data Network Identification Code (DNIC) identify the country?

Data Network Identification Code (DNIC) के kitne digits देश (country) की पहचान करते हैं?

(A) first three / पहले तीन **(B)** first four / पहले चार
(C) first five / पहले पाँच **(D)** first six / पहले छह

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q15] · Hard

Q143. Which of the following is true for Circuit Switching handling multiple simultaneous connections?

एक साथ अनेक connections संभालते समय Circuit Switching के बारे में क्या सही है?

(A) A single path is used and shared for all data / एक ही path सभी data के लिए share होता है **(B)** All connections are multiplexed into a single channel / सभी connections एक channel में multiplex होते हैं
(C) It encodes all data into a single channel / यह सारे data को एक channel में encode करता है **(D)** Separate physical paths are allocated for each channel / हर channel के लिए अलग physical paths दिए जाते हैं

[Source: RPSC Programmer 2024 Paper-1 (with official key)] · Hard

Q144. Which of the following statements is TRUE?

निम्न में से कौन-सा कथन सही है?

(A) The Internet and the WWW are the same thing / Internet और WWW एक ही हैं **(B)** The WWW is a service that runs over the Internet / WWW, Internet पर चलने वाली service है

(C) The Internet is a part of the WWW / Internet, WWW का हिस्सा है **(D)** The WWW does not need the Internet / WWW को Internet की ज़रूरत नहीं

[Source: Practice — original (textbook-based)] · Hard

Answer Key — sirf letters (no solutions)

Q1: A **Q2:** A **Q3:** B **Q4:** C **Q5:** B **Q6:** C **Q7:** D **Q8:** A **Q9:** D **Q10:** C

Q11: B **Q12:** B **Q13:** C **Q14:** A **Q15:** B **Q16:** A **Q17:** D **Q18:** D **Q19:** C **Q20:** C

Q21: B **Q22:** B **Q23:** A **Q24:** A **Q25:** A **Q26:** B **Q27:** A **Q28:** A **Q29:** A **Q30:** C

Q31: C **Q32:** A **Q33:** B **Q34:** C **Q35:** B **Q36:** B **Q37:** C **Q38:** D **Q39:** A **Q40:** A

Q41: B **Q42:** D **Q43:** D **Q44:** A **Q45:** A **Q46:** B **Q47:** A **Q48:** C **Q49:** D **Q50:** B

Q51: D **Q52:** C **Q53:** C **Q54:** B **Q55:** A **Q56:** C **Q57:** C **Q58:** C **Q59:** D **Q60:** C

Q61: B **Q62:** B **Q63:** A **Q64:** C **Q65:** B **Q66:** D **Q67:** B **Q68:** C **Q69:** B **Q70:** D

Q71: C **Q72:** A **Q73:** B **Q74:** D **Q75:** A **Q76:** A **Q77:** C **Q78:** A **Q79:** C **Q80:** B

Q81: A **Q82:** B **Q83:** B **Q84:** A **Q85:** C **Q86:** B **Q87:** C **Q88:** B **Q89:** A **Q90:** C

Q91: B **Q92:** B **Q93:** B **Q94:** C **Q95:** B **Q96:** C **Q97:** B **Q98:** C **Q99:** A **Q100:** A

Q101: B **Q102:** B **Q103:** B **Q104:** D **Q105:** A **Q106:** A **Q107:** C **Q108:** C **Q109:** B **Q110:** C

Q111: D **Q112:** B **Q113:** A **Q114:** C **Q115:** A **Q116:** C **Q117:** D **Q118:** A **Q119:** A **Q120:** B

Q121: C **Q122:** C **Q123:** B **Q124:** A **Q125:** B **Q126:** A **Q127:** A **Q128:** B **Q129:** C **Q130:** A

Q131: A **Q132:** C **Q133:** A **Q134:** D **Q135:** D **Q136:** B **Q137:** B **Q138:** B **Q139:** A **Q140:** C

Q141: D **Q142:** A **Q143:** D **Q144:** B